

By defn. of OR, $\boxed{\neg P(a) \vee \neg R(a)}$ is True.

$\therefore$ For all elements a in the domain, $\boxed{\neg P(a) \vee \neg R(a)}$ is True.

By defn. of universal quantifier, $\forall x (\neg P(x) \vee \neg R(x))$ is True.

By defn. of conditional disjunction equivalence, $\forall x (P(x) \rightarrow \neg R(x))$ is True.

$\therefore$ Conclusion (d) follows from (a), (b) and (c) $\square$

62) Let $P(x)$, $Q(x)$, $R(x)$ and $S(x)$ be the statements "x is a duck", "x is one of my poultry", "x is an officer", and "x is willing to waltz" respectively. Express each of these statements using quantifiers; logical connectives; and $P(x)$, $Q(x)$, $R(x)$ and $S(x)$.

(a) No ducks are willing to waltz: $\forall x (P(x) \rightarrow \neg S(x))$

(b) No officers ever decline to waltz: $\forall x (R(x) \rightarrow S(x))$

(c) All my poultry are ducks: $\forall x (Q(x) \rightarrow P(x))$

(d) My poultry are not officers: $\forall x (Q(x) \rightarrow \neg R(x))$

Does (d) follow from (a), (b) and (c)? If not, is there a correct conclusion?

Let, $\forall x (P(x) \rightarrow \neg S(x))$ be True. By conditional-disjunction equivalence, $\forall x (\neg P(x) \vee \neg S(x))$ is True. By defn. of universal quantifier, for all element a in the domain, $\neg P(a) \vee \neg S(a)$ is True.

Let, $\forall x (R(x) \rightarrow S(x))$ be True. By conditional-disjunction equivalence, $\forall x (\neg R(x) \vee S(x))$ is True. By defn. of universal quantifier, for all elements a in the domain, $\neg R(a) \vee S(a)$ is True.

Let, $\forall x (Q(x) \rightarrow P(x))$ be True. By conditional-disjunction equivalence, $\forall x (\neg Q(x) \vee P(x))$ is True. By defn. of universal quantifier, for all elements a in the domain, $\neg Q(a) \vee P(a)$ is True.

Let, $\neg Q(a)$ be False. Since $\neg Q(a) \vee P(a)$ is True, by defn. of OR, $P(a)$ is True. $\therefore \neg R(a) \vee S(a)$ is True, by defn. of OR, $\neg R(a)$ could either be True or False. Let, it be False. By defn. of negation, $\neg(\neg Q(a))$ is True.